

Risky Behaviour: Which factors contribute to aggressive behaviour and violent offenses in young adults?

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1. Introduction and Problem Definition

Behaviour that negatively impacts a person’s health, well-being, employment opportunities, and social life is considered risky. Smoking, reckless driving, abusing alcohol or other drugs, fighting, and committing crimes are a few examples of risky conduct. This report will specifically examine criminal behaviour, with a particular emphasis on physical violence, to identify which factors contribute to aggressive behaviour and violent offenses in young adults. With serious repercussions for both people and society as a whole, youth violence is still a major social and economic concern. In order to create effective preventative and intervention measures, it is essential to understand the elements that lead to aggressive behaviour during adolescence.

Existing literature on youth violence suggests that multiple factors contribute to aggressive behaviour. This allows us to identify relevant research objectives and hypotheses, by means of comparing our results to previous studies. For example, the research paper by Belley, Castex and Dechter (2023) explores why crime rates among young adults significantly declined from the 1980s to the 2000s in the United States. The authors use data from two National Longitudinal Surveys of Youth (NLSY79 and NLSY97) to examine how possible factors that could lead to youth aggressive behaviour changed over time, dividing them into four categories of interest: biological, social, economic, and the effectiveness of the justice system. Their main finding states that crime participation declined over time, however, it is mentioned that criminal charges increased slightly, so it could be that stricter enforcements are observed, rather than less criminal activity. Moreover, the social factors of the examined time periods prove to be significant, as the number of offenses by adolescents coming from marginalised communities dropped.

However, the research paper published by Belley, Castex and Dechter in 2023 represents one of the few literature pieces that consider the simultaneous analysis of multiple regressors. Although many studies have used the same survey data and answered comparable research objectives, much of the current literature tends to examine a small group of explanatory variables, usually emphasizing one or two regressors or particular interactions. For example, the research paper published by Gottlieb and Sugie (2018) showcases the dependency between marital status and violent behaviour. They divide the relationship in three consecutive stages: entry, stability and exit. This allows them to focus on how changes in a relationship correspond to changes in their criminal conduct over time. In contrast, this report aims to provide a more comprehensive understanding of the determinants of aggressive behaviour, by simultaneously looking at the following five regressors: level of education, marital status, employment history, the respondent’s general health and substance abuse. This wider strategy represents our contribution to the existing literature, as it provides renewed ideas that support and broaden earlier results.

Consequently, utilizing panel data from a longitudinal survey, we apply several econometric models to explore the impact of the five explanatory variables mentioned above. The selected year range is 1998 - 2010, as the survey questions on which regressors are based had complete and consistent data during these years, with no missing values or gaps. Furthermore, the dataset during this time period contains thousands of responses, ensuring a large sample size and adequate information for the research.

The respondents, a sample of US youth, born between 1980-84, are monitored throughout their lives by the NLSY97 longitudinal survey. When the interview was first conducted in 1997, the respondents were between the ages of 12 and 17. Along with a comprehensive collection of observable background factors of the adolescent and their family, a variety of teenage behaviours are surveyed annually from 1997 to 2016.

This survey allows us to use the self-reported responses to the question: “Has the respondent ever attacked someone to hurt or fight?” in order to answer the research question proposed in this report. This measure provides a direct and quantifiable indicator of violent behaviour. However, social desirability bias and underreporting represent two major drawbacks of utilizing self-reported responses to evaluate aggressive behaviour. Given the nature of the question, people can be tempted to give false information out of fear of social or legal repercussions. Consequently, some individuals may conceal or underreport past violent acts, even if they have happened and therefore, this restriction should be taken into account when evaluating the results.

2. Data Description

This analysis uses data from the National Longitudinal Survey of Youth 1997 (NLSY97), which is a panel dataset developed by the U.S. Bureau of Labor Statistics that started in 1997 and is still ongoing. It was created with the purpose of tracking the transition of American youth into adulthood. It includes a variety of different areas, such as demographics, education, employment, family, income, attitudes and criminal activity, with several sub-categories. Thus, it is particularly useful when it comes to analysing violent behaviour over the course of multiple years. This report studies the effects of education, alcohol consumption, health, employment history, and marital status on the likelihood that an individual will attack someone in order to hurt or fight, using data from 1998 up until 2010. For education, the data comes from self-reported responses to the question, “What is the highest grade you have ever completed as of today?”, where individuals could select a number between 0 and 20. This number indicates the completed grade, with 0 meaning “None” and 20 meaning “8th year of college or more.”

To examine the relationship between alcohol use and violent behaviour, the explanatory variable is based on the question, “Number of days in which the respondent has had 5+ drinks in the last 30 days”, referring to a behaviour known as “binge drinking.” Respondents could answer with a number between 0 and 30, representing the number of days of binge drinking, which serves as a satisfactory measure of alcohol abuse.

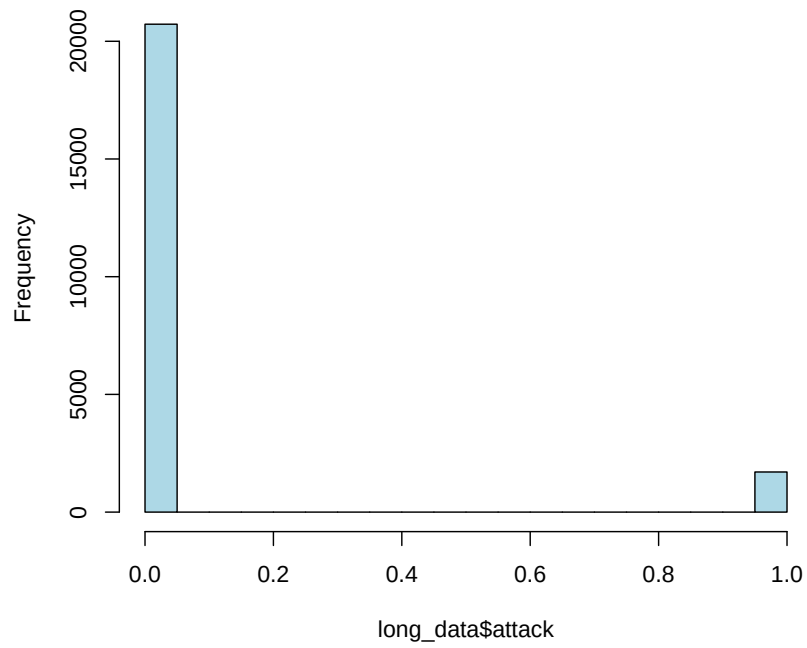
The NLSY97 dataset contains comprehensive information about participants’ health, including the question, “How is the respondent’s general health?”. The respondents’ answers were assigned numerical values: 5 for Poor, 4 for Fair, 3 for Good, 2 for Very Good, and 1 for Excellent. These values allow the applied models to explain the effect of health on violent behaviour or attacks. The econometric models used in this report quantify employment based on self-reported responses to the question, “Number of weeks the respondent has worked at any employee-type job during the year,” with answers ranging from 1 to 53 weeks.

For marital status, the selected question is, “What is the respondent’s current marital status?” with responses falling into the following categories: 0 for Never-married, 1 for Married, 2 for Separated, 3 for Divorced, and 4 for Widowed. Given the nature of the responses, this report treats marital status as a categorical variable, with Never-married as the baseline category.

The data for the sex of the respondents was recorded in 1997 and is 1 for Female and 0 for Male. The race/ethnicity of the respondents was also recorded in 1997 and has the following categories: 1 for Black, 2 for Hispanic, 3 for Mixed Race (Non-Hispanic), and 4 for Non-Black /Non-Hispanic.

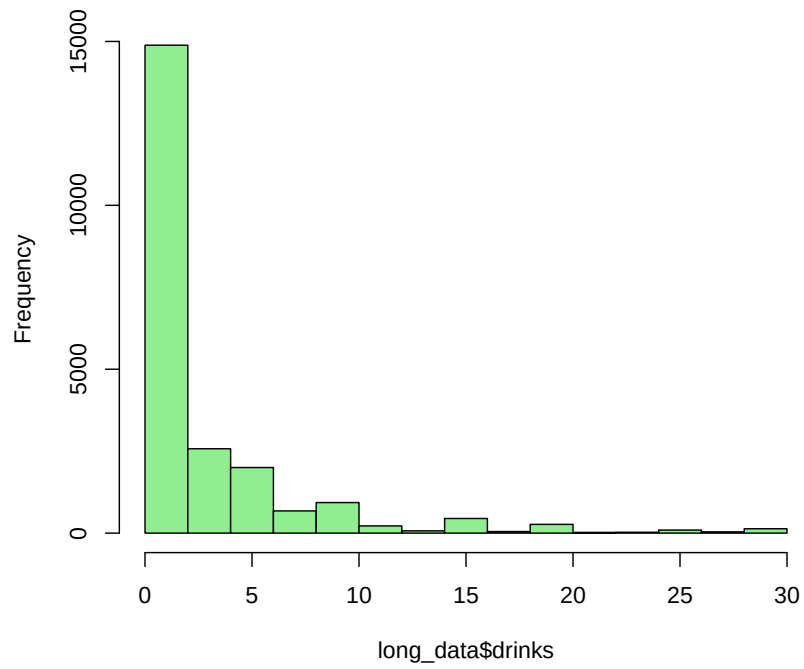
The histograms and descriptive statistics found in Table 1 show that most of the individuals in the sample have not reported violent behaviour, with only 8% indicating an attack. This can also be concluded from the histogram of Attack, which displays a sharp spike at the zero value. Given the 1-5 scale on which health is measured, the respondents consider their health to be very good on average, as the variable “health” has a mean of 2.12. The level of education clusters around a mean of 12.43 years, which indicates that the majority of participants have at least a high-school diploma. Most people reported low levels of alcohol consumption, with an average of 3 days in which they have consumed five or more drinks. However, there are a few noticeable outliers that drive the maximum of this variable to 30, revealing that some respondents engaged in heavy drinking. A sizeable portion of individuals are fully employed, as denoted by the spike at the value of 53 weeks of employment, but there is considerable variation in the levels of work participation, with the standard deviation being approximately 20 weeks. Overall, these results suggest that in general the population is nonviolent, healthy, educated, with slight variations when it comes to employment status and alcohol consumption.

Figure 1. Frequency of Respondents Attacking Someone



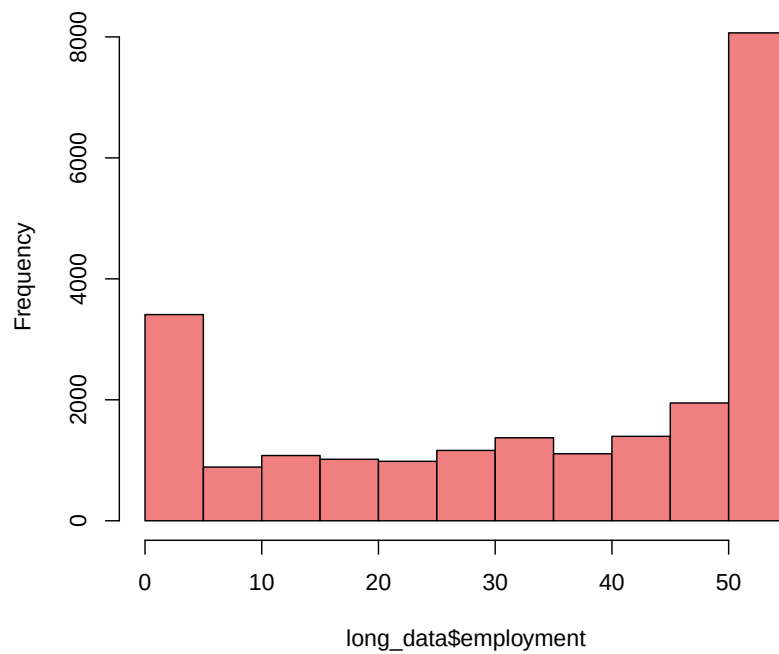
Note: 1 = respondent attacked someone to hurt or fight. This shows the frequency of such behavior between 1990–2010. Most respondents did not engage in this behavior.

Figure 2. Frequency of Binge Drinking



Note: Number of days with 5+ drinks per month. Displays how often respondents binge drank.

Figure 3. Frequency of Employment (Weeks Worked)



Note: Total weeks worked per year. Shows how many respondents worked specific numbers of weeks.

3. Methods

This study proposes the use of four different models to analyse the prevalence of U.S. individuals towards committing violent crimes. The models vary in complexity, purpose of use and fitness to the data, providing a range of perspectives and insights relevant to answering the research question.

We introduce the following notation used for describing the models:

- N : number of individuals in the sample,
- T : number of time periods in the panel,
- y_{it} : value of binary dependent variable for individual i in time period t ,
- $x_{it} = [\text{health}_{it}, \text{education}_{it}, \text{marstatMarried}_{it}, \text{marstatSeparated}_{it}, \text{marstatDivorced}_{it}, \text{marstatWidowed}_{it}, \text{drinks}_{it}, \text{employment}_{it}]'$: column vector containing the values of the explanatory variables for individual i in time period t ,
- $\beta = [\beta_1, \beta_2, \beta_3, \beta_4, \beta_5, \beta_6, \beta_7, \beta_8]'$: column vector containing the coefficients corresponding to each regressor,
- β_0 : intercept (constant term),
- ε_{it} : error term for individual i in time period t .

3.1 Pooled Ordinary Least Squares (OLS) Model

A linear pooled OLS model is applied to the data to develop an initial holistic view of the effects of different independent variables. The available data for each individual and each year is pooled and the regression equation is given by:

$$y_{it} = \beta_0 + (x_{it})'\beta + \varepsilon_{it}, \quad i = 1, \dots, N, \quad t = 1, \dots, T. \quad (1)$$

As Cameron and Trivedi (2005) explained, if the exogeneity assumption $Cov(\varepsilon_{it}, x_{it}) = 0$ is satisfied, then the pooled OLS estimator is consistent. However, if the true model contains fixed effects, then the pooled OLS model would yield inconsistent results. In this study, we are likely faced with the latter case. Due to the panel structure of the data, omitted variable bias caused by unobserved heterogeneity that is constant over time might be present. In other words, we should control for time-invariant unobserved characteristics that are specific for each individual by introducing individual-specific effects in the model. An example of unobserved heterogeneity could be one's temperament, which can be assumed to not vary significantly over time and was suggested to be an important determinant of violent behaviour in adolescence (Kutlllovci et al., 2022).

Furthermore, a fixed effects model would be preferred to a random effects model because of the likely correlation between the individual-specific effects and the regressors. For instance, several studies suggested the existence of a correlation between gender and alcohol consumption (Wilsnack et. al., 2009; Dawson & Archer, 1992). Since using a random effects model when fixed effects are present would yield inconsistent results (Wooldridge, 2006), fixed effects would be more suitable in our context. To formally verify the presence of fixed effects, the test proposed by Hausman (1978) was used. The Hausman test checks whether the coefficients of time-varying regressors are significantly different when using random effects and fixed effects estimation, respectively. Under the null hypothesis, their difference is statistically insignificant and random effects may be used. However, if the null hypothesis is rejected, the Hausman test indicates the presence of fixed effects, hence the fixed effects estimator should be used.

While the pooled OLS model serves as a simple starting point in exploring the relationships within the data, it fails to exploit the panel data structure and to offer reliable results, due to inconsistency caused by the likely presence of fixed effects.

3.2 Pooled OLS with Fixed Effects Model

We subsequently consider a model more suitable for the analysis of panel data. Since observations are not independent across time, it is relevant to consider individual-specific effects in the form of fixed effects, as explained in section 2.1. Thus, we alter the pooled OLS model in (1) by adding individual-specific effects:

$$y_{it} = \alpha_i + (x_{it})' \beta + \varepsilon_{it}, \quad i = 1, \dots, N, \quad t = 1, \dots, T,$$

where α_i represents the individual-specific effects measuring unobserved heterogeneity, which can potentially be correlated with the regressors (Cameron & Trivedi, 2005). To estimate this model, we make the following assumptions as presented by Cameron and Trivedi (2005): ε_{it} are independent and identically distributed over i and t and strict exogeneity holds (i.e. $E[\varepsilon_{it} | \alpha_i, x_{i1}, \dots, x_{iT}] = 0$, $t = 1, \dots, T$).

The pooled OLS model with fixed effects still poses substantial limitations. We recall that the data contains a binary outcome variable y_{it} . The current model neglects the discreteness of the dependent variable due to the linearity of the model, as the fitted values do not necessarily lie between 0 and 1 (Cameron & Trivedi, 2005). This may lead to predictions that are outside the range of possible outcomes, which is undesirable. Thus, this considerable limitation suggests the need for other, more suitable models for binary outcome data.

3.3 Fixed Effects (FE) Logit Model

After the estimation of OLS models, a fixed effects logit model was studied. The model employs maximum likelihood estimation and, as presented by Chamberlain (1980), is given by:

$$Pr(y_{it} = 1 | x_{it}, \beta, \alpha_i) = F(\alpha_i + (x_{it})' \beta), \quad (2)$$

where F represents the cumulative distribution function of the standard logistic distribution:

$$F(z) = \frac{e^z}{1 + e^z}, \quad z \in \mathbb{R}.$$

The distinctive features of this nonlinear model address both the panel structure of the data, as well as the discreteness of the dependent variable. The logit model ensures that $Pr(y_{it} = 1 | x, \beta, \alpha)$ attains values strictly between 0 and 1, for any x , β and α . This is achieved through the use of the logistic function F , making the logit model suitable for binary responses (Wooldridge, 2006). Moreover, the individual-specific effects α_i account for unobserved heterogeneity that is typical for panel data, while they are allowed to be correlated with the regressors through the use of fixed effects.

We subsequently proposed two extensions of the FE logit model. In the first extension, year fixed effects were added by introducing a set dummy variables corresponding to each year as regressors in the model described in (2). To avoid multicollinearity, the dummy corresponding to the reference year 1998, the earliest year in the data set, was omitted. Year fixed effects were added to account for potential time-varying unobserved factors that affect all individuals in a specific year, which include major U.S. policy changes and macroeconomic shocks. For example, it is possibly relevant to control for the effects of the financial crisis of 2008 because it might drive changes in the employment of individuals and 2008 is a year belonging to the panel data's time frame. The extended model using year fixed effects is hence given by:

$$Pr(y_{it} = 1 | x_{it}, \beta, \alpha_i, \theta_t) = F(\alpha_i + \theta_t + (x_{it})' \beta),$$

where θ_t represents the year fixed effects.

Another extension of the FE logit model consists of including a linear time trend. This was done by introducing the year as an additional regressor in the model expressed in (2). This yields:

$$Pr(y_{it} = 1 | x_{it}, \beta, \alpha_i) = F(\alpha_i + (x_{it})' \beta + \beta_{year} year),$$

where $year$ is an explanatory variable storing the year and β_{year} represents its corresponding coefficient. The reason for estimating such a model lies in the possibility of an existing time trend, that is, to test whether the frequency of violent crimes systematically changes over time as a result of years passing.

Nevertheless, the FE logit model has several limitations. Notably, the individuals presenting no variation across time in the values of the dependent variable drop out of the analysis. The resulting loss of data, consisting of 17,531 removed observations, represents a trade-off for eliminating the individual-specific effects in the regression. Another significant disadvantage of fixed effects consists of not allowing the estimation of coefficients of time-invariant regressors, as these are absorbed into the individual-specific effects (Cameron & Trivedi, 2005). Therefore, this provides motivation to explore more advanced models to be able to estimate effects of factors such as gender and ethnicity.

3.4 Correlated Random Effects (CRE) Model

The CRE approach brings together the advantages of both fixed effects and random effects models, as it is possible to estimate the effects of time-invariant factors, while allowing the individual-specific effects to be correlated with the regressors. This is done by applying the random effects estimation method to a model that includes the panel averages of regressors that vary over time as additional independent variables.

Mundlak (1978) showed that, for a linear model, the obtained CRE coefficient estimates for time-varying regressors are the same as the fixed effects estimates. This result, however, was limited to balanced panels until Wooldridge (2019) extended it to unbalanced panels as well, considerably increasing its applicability. Furthermore, Wooldridge (2019) went beyond the linear CRE model and gave several indications for possible CRE approaches for nonlinear models as well. The application of the CRE method to nonlinear models for unbalanced panels was further developed by Bates et al. (2022).

In what follows, we adapted the linear CRE approach as described in Wooldridge (2006, Chapter 14) to a nonlinear CRE model using the logistic distribution to estimate the effects of gender and ethnicity on the dependent variable. For this purpose, we introduce the following additional notation:

α_i : random variable containing individual-specific effects for individual i ,

$\bar{x}_i = T^{-1} \sum_{t=1}^T x_{it}$: column vector containing the time averages of the time-varying regressors for individual i ,

$z_i = [gender_i, ethnicity_i]'$: column vector containing the values of the time-invariant regressors for individual i ,

$\delta = [\delta_1, \delta_2]'$: column vector containing the coefficients corresponding to the time-invariant regressors.

Assume the following linear relationship:

$$\alpha_i = a + \gamma \bar{x}_i + r_i, \quad (3)$$

where the error term r_i is uncorrelated to any x_{it} such that: $Cov(\bar{x}_i, r_i) = 0$. Adapting the model in (2) by adding the time-invariant regressors z_i and rewriting the random individual-specific effects α_i using (3) yields the following CRE model:

$$Pr(y_{it} = 1 | x_{it}, z_i, \beta, \delta, a, \gamma, r_i) = F(a + (x_{it})' \beta + \bar{x}_i' \gamma + z_i' \delta + r_i). \quad (4)$$

Since the equality between the CRE and FE coefficients of time-varying regressors holds in the case of linear models (Mundlak, 1978; Wooldridge, 2019) but does not generally hold for nonlinear models such as logit, it is relevant in our study to interpret the estimates of δ , as well as of β as presented in (4).

4. Results

4.1 Pooled Ordinary Least Squares (OLS) Model

The first model applied to the data is the pooled OLS model. As seen in Table 2, the results highlight that health status, education, alcohol consumption, employment, and marital status, specifically being married, are all highly statistically significant predictors of violent behaviour, with p -values equal to 0.0000. The other regressors, namely the marital statuses “Separated”, “Divorced” and “Widowed” are not statistically significant, as they have p -values higher than 0.1. The intercept of 0.24 reflects the predicted probability of violent behaviour for an individual in the baseline group, i.e. someone who has never been married and is in the reference category for all other variables. Given that health is measured on a scale from 1 (excellent) to 5 (poor), the coefficient on health suggests that as health worsens by one unit, the probability of attacking someone increases by 1.06 percentage points, holding other factors constant. Education has a negative impact on the likelihood of attacking: each additional year of education is associated with a 1.48 percentage point decrease in the probability of committing violent acts, *ceteris paribus*. Controlling for other variables, each additional day per month in which an individual consumes more than five alcoholic drinks is linked to a 0.64 percentage point increase in the probability of engaging in violence. Moreover, each additional week of employment during the year is associated with a 0.07 percentage point decrease in the likelihood of attacking someone, keeping changes in the other factors fixed. Finally, married individuals are 2.79 percentage points less likely to commit violence compared to people who have never been married, *ceteris paribus*.

The Hausman test is then conducted in order to ascertain whether fixed or random effects should be used. The obtained p -value is less than 0.0001, which leads to the rejection of the null hypothesis. The conclusion of the Hausman test is that the random effects model is inconsistent, and the fixed effects model is preferred for this data set.

4.2 Pooled OLS with Fixed Effects Model

Applying fixed effects to the pooled OLS model results in some changes compared to the pooled OLS with no effects (see Table 3). The variables for education, alcohol consumption, employment, and “Married” marital status remain statistically significant, but health is no longer significant. Compared to the OLS model with no effects, the coefficients for education, alcohol consumption and employment slightly decrease, while the coefficient for marital status “Married” increases. The marital status “Divorced” now has a p -value of 0.0000, making it highly statistically significant. In this model, each additional year of education is associated with a 1.27 percentage point decrease in the probability of committing violent acts, holding other factors constant. Controlling for the other variables, each additional day per month of consuming more than five alcoholic drinks is linked to a 0.28 percentage point increase in the likelihood of engaging in violence. Moreover, each additional week of employment during the year is associated with a 0.04 percentage point decrease in the likelihood of attacking someone, keeping changes in the other factors fixed. Married individuals are 7.1 percentage points less likely to commit violence compared to those who have never been married, while the probability for divorced individuals is 12.5 percentage points lower, *ceteris paribus*.

4.3 Fixed Effects Logit Model

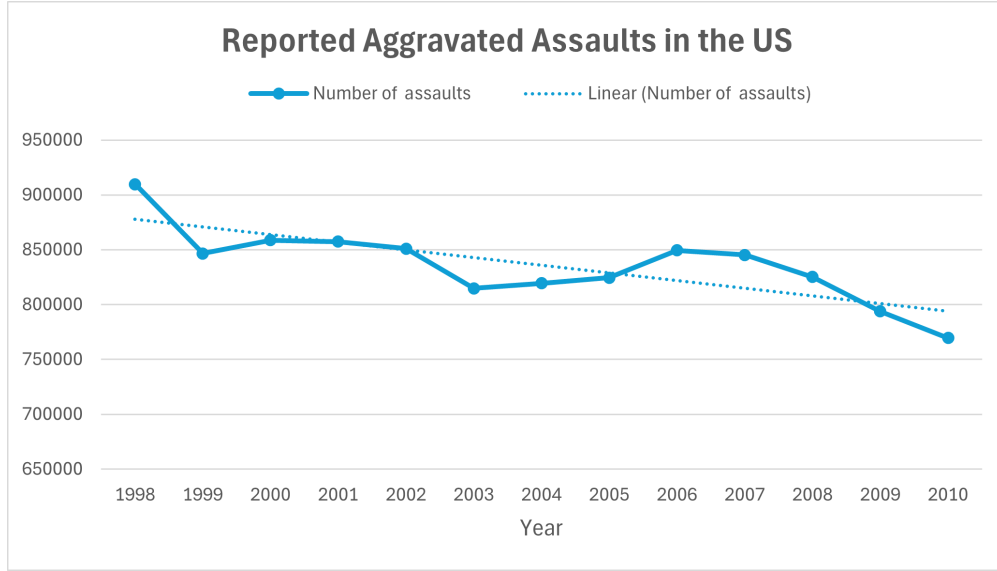
After obtaining a set of initial results from the OLS models, a logit model with fixed effects was applied to the data, with results found in Table 4. This approach indicated that level of education, alcohol consumption and marital status are highly significant predictors of one’s violent behaviour, with p – value = 0.0000. In particular, an additional day per month in which an individual consumes a minimum of five alcoholic drinks corresponds to an increase in the chances of them committing violent crimes by a factor of $e^{0.043465} \approx 1.0444$, when accounting for the changes in their health, education, marital status and employment over time.

Contrastingly, an additional grade of education completed is associated with a chance lower by a factor of $e^{-0.273175} \approx 0.7610$ of exhibiting violent behaviour within the same individual, after controlling for the other individual factors. In terms of marital status, an individual becoming married is associated with decreased odds of carrying out a violent attack by a factor of $e^{-1.595846} \approx 0.2027$ as compared to the odds corresponding to the same individual when they were never married, holding the changes in all other observed factors fixed. Moreover, employment was shown to be a significant factor in predicting violent crimes at 0.1% significance level, being negatively correlated with prevalence towards violence. The effects of employment are the most precise estimates of this analysis, as the standard errors are the smallest among the regressors. As opposed to the pooled OLS model, the fixed effects logit model suggests that one's general health is statistically insignificant in explaining the likeliness of engaging in violent actions.

Furthermore, year fixed effects were added to the model. This resulted in alcohol consumption and marital status being the only statistically significant factors, with high significance. The results suggest that, controlling not only for individual changes in the observed factors but also for unobserved year-specific influences, an additional day per month of considerable alcohol consumption corresponds to an increase in the chances of a violent attack by a factor of $e^{0.048705} \approx 1.0499$. Since this value is further away from 1 than the factor estimated by the logit model without year fixed effects, the current model suggests a slightly more powerful impact of drinking. Thus, alcohol represents a significant predictor of violent tendencies even after accounting for macro-level policy changes in the United States between 1998 and 2010 and macroeconomic aspects, such as the financial crisis of 2008. When it comes to marital status, an individual becoming married is associated with decreased odds of engaging in a violent attack by a factor of $e^{-0.844385} \approx 0.4298$ as compared to the odds corresponding to the same individual when they were never married, keeping the changes in all other observed factors constant. This factor is considerably closer to 1 than the factor estimated by the initial fixed effects logit model, which implies that, when year fixed effects are included, the effects of marital status are estimated to be less significant. This is confirmed by the decrease in p -values as displayed in Table 4 and Table 5 in the Appendix.

Subsequently, a time trend was added to the initial fixed effects logit model. Similar to the previous model including year fixed effects, alcohol consumption and marital status constituted the only statistically significant factors among the original regressors, with high significance levels and coefficients similar to those estimated by the model with year fixed effects. Additionally, the year itself represented another highly significant factor with p -value = 0.0000, indicating the presence of a time trend in which later years are associated with a systematic decline in violent attacks. This result is in agreement with U.S. crime data provided by the Federal Bureau of Investigation (FBI) (2025), which shows a declining trend in the number of reported assaults from 1998 to 2010, as displayed in Figure 4.

Figure 4. Reported aggravated assaults in the U.S. between 1998-2010



4.4. Correlated Random Effects (CRE) Model

After having completed the Logit model, the CRE (Correlated Random Effects) Model was used. Unlike the logit model, this model estimates the time-invariant variables such as sex and race. In order to control for correlation between regressors and individual-specific effects, the mean of each regressor except for marital status (since it is categorical) was included in the model. These new regressors estimate the effects between individuals, while the initial regressors estimate the effects within an individual over time similar to the fixed effects logit model. This model indicated the most significant predictors (at a significance level of 0.1%) of violent behaviour are education, marital status, alcohol consumption, race, and sex. A slightly less significant regressor is employment, while the regressors that are not significant is health. According to the model, individuals who “binge drink” more over time, are significantly more likely to exhibit violent behaviour. A one day increase in an individual’s average monthly “binge drinking” leads to a $e^{0.0932}$ (9.8%) increase in the odds of attacking someone. When a person is mostly married over time, the odds of attacking someone also increase by $e^{1.016}$ versus when they are unmarried over time. This is contrary to the CRE model’s indication that individuals are less likely to commit violence in the years they are married versus in the years they have not been married before. For the time-invariant variables, sex and race, the model indicated that the odds of attacking someone decrease by $e^{0.5583}$ if someone is female, and decrease by $e^{0.5583}$ if someone is Non-Black / Non-Hispanic relative to the base case (race is Black) holding all other factors constant.

5. Discussion

Using individual-level data and four econometric models: pooled OLS, OLS with fixed effects, a correlated random effects (CRE) specification, and a fixed effects logit model, this paper looked at the factors influencing violent behaviour, defined as attacking someone to hurt or fight. Overall, the fixed effects models underlined the importance of considering unobserved time-invariant traits when looking at relevant societal or economic factors, because, for instance, natural aggression or early-life environment could distort initial findings. Our quantitative findings align with the results obtained by various other existing papers, thus providing additional insights into which factors determine individuals to engage in aggressive conduct.

The highest grade completed proved to be highly statistically significant in all four models used in this report. According to our results, each additional grade level completed reduces the probability of attacking someone by 0.0127 units in the OLS model with fixed effects, or by 0.0148 units in the classical pooled OLS model. This is not a surprising result, because research shows that the probability of engaging in physical violence as a young adult increases amongst the individuals with a lower level of completed education. By improving conflict resolution and emotional control, education lessens impulsive and violent reactions. This happens due to the controlled settings that schools offer, which deter violence and promote social behaviour (Lochner, 2020).

Another regressor that appears to be highly statistically significant in all four models is alcohol use. In the fixed effects logit model, an additional day of binge drinking is associated with a 4.44% increase in the likelihood of attacking someone. In a similar manner, Chrusciel examines, in a research paper from 2020, the relationship between alcohol use and criminal behaviour, also using the NLSY97 panel data and the number of binge drinking days in a month. The findings show that a one day increase in this number increases the probability of arrest by 7%. The study conducted by Popovici et al. in 2012 also suggests that alcohol use is strongly associated with both committing and being a victim of crime, especially among young adults. However, their study claims that property crimes, such as theft, vandalism and intrusion are more strongly associated with drinking than violent crimes, such as attacking or fighting.

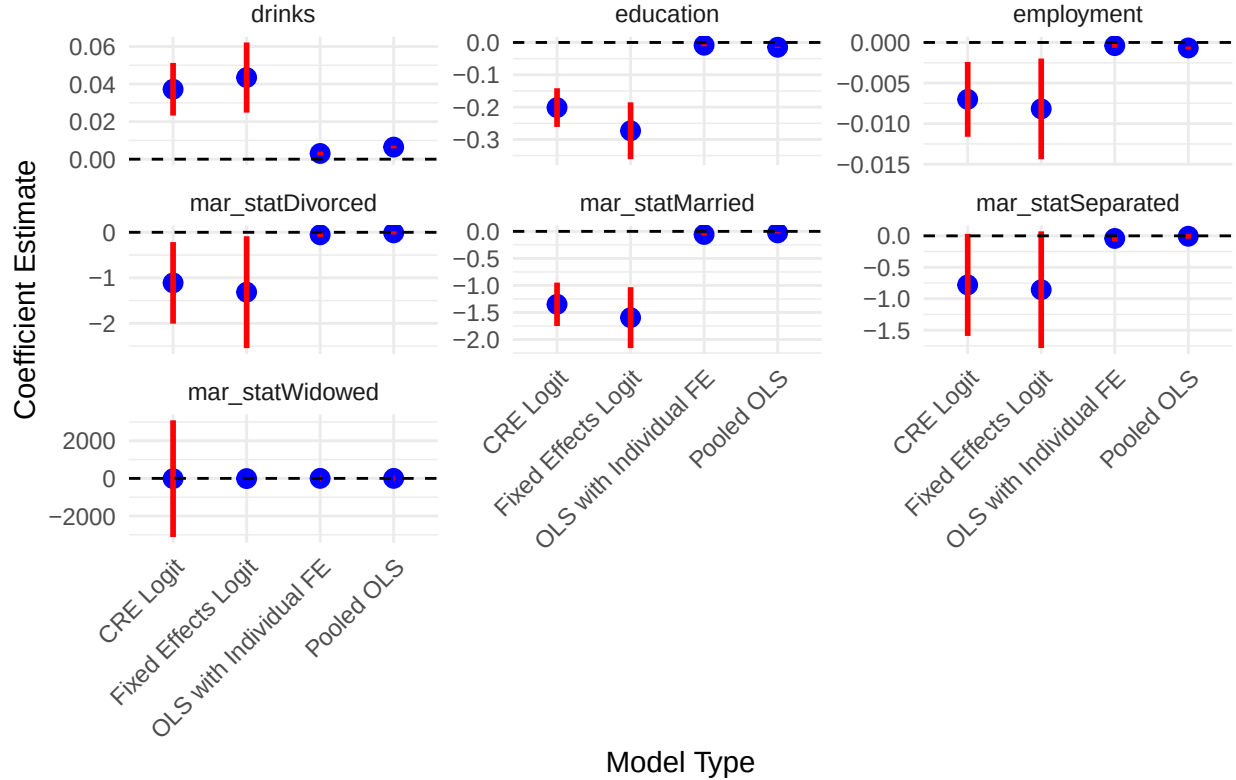
In the case of examining marital status, the marginal effect of the category “Married” relative to the never-married baseline appeared to be highly statistically significant in all four models. We see that the OLS model with fixed effects associated being married, as opposed to never-married, with a 0.0711 units decrease in the probability of attacking someone, all else equal, whereas in the general pooled OLS model, married individuals have a 0.02792 units lower predicted attack rate versus the never-married group. In a slightly different manner, but using the same dataset, the research paper published by Gottlieb and Sugie in 2018 showcases the dependency between marital status and violent behaviour. Their findings show that while both marriage and cohabitation lead to diminished criminal behaviour, marriage has a more significant impact. Generally, individuals cohabitating were 13% less likely to engage in violent behaviour, as opposed to their single counterparts, and married ones were 35% less likely. More specifically, stable cohabitation reduced the crime probability by 29% and stable marriage by 55%.

At the same time, only the OLS with fixed effects model did not find employment history as a statistically significant regressor. The pooled OLS states that an additional week worked is associated with a 0.006786 units decrease in the probability of attacking someone to hurt or fight, while the fixed effects logit model associates each additional week worked with a 0.82% decrease. Although the study conducted by Apel and Horney in 2017 focuses on the employment situation and its changes from one year to another, their findings are still notable: young people who have a longer history of steady work are considerably less likely to commit violent crimes. In contrast, a higher likelihood of violence is associated with unemployment, precarious employment, or irregular work history. Additionally, they propose that employment serves as a protective factor by offering financial stability, structure, and a decreased exposure to surroundings that are favourable to criminal activity. The key findings state that a one-year increase in work experience reduces the likelihood of violent behaviour by 1.3 to 1.9 percentage points. On the other hand, an additional year spent in unemployment raises the probability of violence by about 1.5 percentage points.

Lastly, the general health of the individual appears to be statistically significant only in the pooled OLS model and in the correlated random effects logit model. In the pooled OLS model, better health is associated with 0.01057 units increase in the probability of attacking, holding other factors constant. In contrast, a research paper published by Otsu and Yuen in 2022 explores how the general health of an individual impacts the likelihood of engaging in physical violence, in particular attacking or hurting someone on purpose. Using the NLSY97 panel data and the fixed effects logit model, they discover a clear correlation between poorer general health and a higher risk of violent behaviour: individuals with “poor” overall health were 3.5 times more likely to report participating in violent behaviour than those with “excellent” health.

These patterns are also reflected in the following coefficient plots, which visually compare the estimated effects and confidence intervals across models for each regressor.

Figure 5. 95% Confidence Intervals Across Models by Regressor



6. Conclusion

In conclusion, these results open the path towards two possible future courses of action. Firstly, we consider that future research papers could focus on the three most significant factors identified in this paper: level of education, marital status and binge drinking. However, it is important to research beyond their individual effects on the likelihood of violent behaviour, and include possible interactions between them.

Moreover, our results underline the urgency of developing targeted social policies. By improving access to educational opportunities and decision-making abilities among young adults, particularly in areas with historically poor achievement, we could observe decreases in aggressive behaviour. Additionally, programs that support young adults in finding stability in relationships and minimize the obstacles associated with marriage could help to indirectly cut violence rates. Most importantly, initiatives that aim to reduce the number of violent crimes should prioritize substance abuse prevention programs as one of the fundamental elements.

Consequently, governments and community oriented organisations can effectively interpret the significance of different factors in determining individuals to engage in attacks. We suggest using the evidence provided in this paper, alongside past academic research published on this topic, in order to create policies that directly address the identified high risk factors.

7. References

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8. Appendix

attack – Name of the binary dependent variable that takes value 1 if an individual has attacked someone to hurt or fight, and 0 otherwise
 drinks – Variable that measures the number of days in which an individual has had 5+ drinks/month
 education – Variable that measures the number of years of education from 0 (none) to 20 (8th year of college or more)
 employment – Variable that measures the number of weeks worked in a year, from 0 to 53
 health – Variable that measures the health of the individual on a scale from 1 (Excellent) to 5 (Poor)
 mar_statDivorced – Binary variable that takes value 1 if the individual is divorced, 0 otherwise
 mar_statMarried – Binary variable that takes value 1 if the individual is married, 0 otherwise
 mar_statSeparated – Binary variable that takes value 1 if the individual is separated, 0 otherwise
 mar_statWidowed – Binary variable that takes value 1 if the individual is widowed, 0 otherwise
 mean_drinks – The mean of the drinks variable
 mean_education – The mean of the education variable
 mean_employment – The mean of the employment variable
 mean_health – The mean of the health variable
 mean_divorced – The mean of the mar_statDivorced variable
 mean_married – The mean of the mar_statMarried variable
 mean_separated – The mean of the mar_statSeparated variable
 mean_widowed – The mean of the mar_statWidowed variable
 raceHispanic – Binary variable that takes value 1 if the individual is Hispanic, 0 otherwise
 raceMixed Race (Non-Hispanic) – Binary variable that takes value 1 if the individual is mixed Race (non-Hispanic), 0 otherwise
 raceNon-Black / Non-Hispanic – Binary variable that takes value 1 if the individual is not black or Hispanic, 0 otherwise
 sex – Binary variable that takes value 1 if the individual is female, 0 if male
 year – The year itself, used in the time trend

. denotes significant at the 10% level, * denotes significant at the 5% level, ** denotes significant at the 1% level, *** denotes highly significant at the 0.1% level

Table 1: Descriptive Statistics

variable	mean	sd	min	max
attack	0.08	0.26	0	1
health	2.12	0.93	1	5
education	12.43	2.00	0	20
drinks	2.99	4.84	0	30
employment	33.79	19.45	0	53

Table 2: Pooled OLS

Variable	Estimate	Std_Error	t_value	p_value	Significance
(Intercept)	0.2439000	0.0124100	19.651	0.0000	***
health	0.0105700	0.0018880	5.596	0.0000	***
education	-0.0148000	0.0008943	-16.551	0.0000	***
drinks	0.0064270	0.0003600	17.853	0.0000	***
mar_statMarried	-0.0279200	0.0062360	-4.478	0.0000	***
mar_statSeparated	-0.0065430	0.0197800	-0.331	0.741	

Variable	Estimate	Std_Error	t_value	p_value	Significance
mar_statDivorced	-0.0126300	0.0168400	-0.750	0.453	
mar_statWidowed	-0.0908000	0.0984200	-0.923	0.356	
employment	-0.0006786	0.0000909	-7.468	0.0000	***

Note: Pooled OLS regression results estimating the probability of attacking someone as a function of health, education, alcohol consumption, employment, and marital status - with categories: never-married (baseline), married, separated, divorced, and widowed.

Table 3: Pooled OLS with Fixed Effects Model

Variable	Estimate	Std_Error	t_value	p_value	Significance
health	0.0001898	0.0028980	0.0655	0.9477700	
education	-0.0085405	0.0016165	-5.2833	0.0000001	***
drinks	0.0030530	0.0004978	6.1324	0.0000000	***
mar_statMarried	-0.0572212	0.0097514	-5.8680	0.0000000	***
mar_statSeparated	-0.0411480	0.0248186	-1.6580	0.0973500	.
mar_statDivorced	-0.0546422	0.0236920	-2.3064	0.0211000	*
mar_statWidowed	-0.0762709	0.1386806	-0.5500	0.5823400	
employment	-0.0003930	0.0001364	-2.8817	0.0039600	**

Note: Pooled OLS with Fixed Effects regression results estimating the probability of attacking someone as a function of health, education, alcohol consumption, employment, and marital status - with categories: never-married (baseline), married, separated, divorced, and widowed.

Table 4: Fixed Effects Logit Model Summary

Variable	Estimate	Std_Error	z_value	p_value	Significance
health	0.006489	0.060597	0.107083	0.9147200	
education	-0.273175	0.044886	-6.085934	0.0000000	***
mar_statMarried	-1.595846	0.287301	-5.554605	0.0000000	***
mar_statSeparated	-0.857428	0.473865	-1.809433	0.0703840	
mar_statDivorced	-1.315500	0.626849	-2.098591	0.0358530	*
mar_statWidowed	-10.676753	0.357852	-29.835677	0.0000000	***
drinks	0.043465	0.009530	4.561077	0.0000051	***
employment	-0.008184	0.003163	-2.587677	0.0096626	**

Note: Fixed Effects Logit regression results estimating the probability of attacking someone as a function of health, education, alcohol consumption, employment, and marital status - with categories: never-married (baseline), married, separated, divorced, and widowed.

Table 5: Fixed Effects Logit Model (with Year) Summary

Variable	Estimate	Std_Error	z_value	p_value	Significance
health	0.050699	0.063648	0.796552	0.4257100	
education	0.005773	0.048763	0.118384	0.9057600	
mar_statMarried	-0.844385	0.288755	-2.924231	0.0034531	**
mar_statSeparated	0.110003	0.448996	0.244998	0.8064600	
mar_statDivorced	-0.224746	0.605622	-0.371099	0.7105600	

Variable	Estimate	Std_Error	z_value	p_value	Significance
mar_statWidowed	-8.680686	0.448735	-19.344787	0.0000000	***
drinks	0.048705	0.010319	4.720123	0.0000024	***
employment	-0.003227	0.003342	-0.965545	0.3342700	

Note: Fixed Effects Logit (with year fixed effects) regression results estimating the probability of attacking someone as a function of health, education, alcohol consumption, employment, and marital status - with categories: never-married (baseline), married, separated, divorced, and widowed.

Table 6: Fixed Effects Logit Model (with Time Trend) Summary

Variable	Estimate	Std_Error	z_value	p_value	Significance
health	0.056691	0.063412	0.894009	0.3713200	
education	-0.019246	0.047593	-0.404401	0.6859200	
mar_statMarried	-0.844586	0.294054	-2.872220	0.0040760	**
mar_statSeparated	0.056294	0.460182	0.122329	0.9026400	
mar_statDivorced	-0.163272	0.614889	-0.265531	0.7906000	
mar_statWidowed	-8.535921	0.444575	-19.200182	0.0000000	***
drinks	0.046876	0.010219	4.587141	0.0000045	***
employment	-0.003978	0.003310	-1.201977	0.2293700	
year	-0.260509	0.026235	-9.929898	0.0000000	***

Note: Fixed Effects Logit (with time trend) regression results estimating the probability of attacking someone as a function of health, education, alcohol consumption, employment, and marital status - with categories: never-married (baseline), married, separated, divorced, and widowed.

Table 7: Summary of CRE Logit Model

Variable	Estimate	Std. Error	z value	Pr(> z)	Significance
(Intercept)	-0.5638	0.3758	-1.500	0.1335	
health	0.0049	0.0489	0.101	0.9199	
education	-0.2015	0.0305	-6.604	0.0000	***
drinks	0.0372	0.0072	5.208	0.0000	***
employment	-0.0070	0.0024	-2.982	0.0029	**
mar_statMarried	-1.3490	0.2047	-6.590	0.0000	***
mar_statSeparated	-0.7806	0.4142	-1.884	0.0595	.
mar_statDivorced	-1.1110	0.4572	-2.430	0.0151	*
mar_statWidowed	-13.4500	1411.0000	-0.010	0.9924	
mean_health	0.3428	0.0758	4.520	0.0000	***
mean_education	-0.0759	0.0400	-1.897	0.0578	.
mean_drinks	0.0932	0.0137	6.828	0.0000	***
mean_employment	-0.0046	0.0036	-1.277	0.2015	
mean_married	1.0160	0.2694	3.772	0.0002	***
mean_separated	0.8355	0.7986	1.046	0.2955	
mean_divorced	1.6380	0.7132	2.296	0.0217	*
mean_widowed	-3.4770	9.0060	-0.386	0.6994	
sex	-0.5583	0.0949	-5.882	0.0000	***
raceHispanic	-0.4058	0.1301	-3.120	0.0018	**
raceMixed Race (Non-Hispanic)	-0.8788	0.4800	-1.831	0.0671	.

Variable	Estimate	Std. Error	z value	Pr(> z)	Significance
raceNon-Black / Non-Hispanic	-0.6938	0.1112	-6.241	0.0000	***

Note: Correlated Random Effects regression results estimating the probability of attacking someone as a function of health, education, alcohol consumption, employment, and marital status - with categories: never-married (baseline), married, separated, divorced, and widowed, demographics (sex; race - Hispanic, mixed race (non-Hispanic), non-black), and the means of health, education, alcohol consumption and employment.